AERM Exam Outputs and Results

2026-03-23

Part 1

Use Julia code

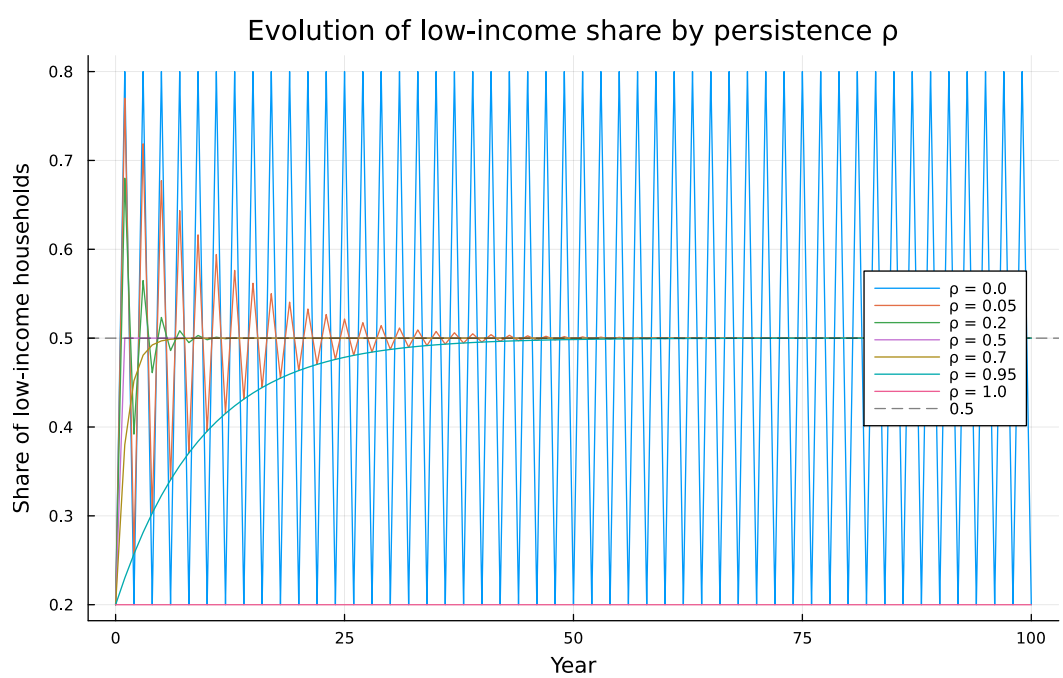
Exercise 1: Solve Standard Consumption-Saving Problem for Different Values of ρ

Q1: Stationary Income Distribution

Income follows a two-state Markov chain with symmetric persistence $\rho_L = \rho_H = \rho$ and transition matrix:

$$P = \begin{pmatrix} \rho & 1-\rho \\ 1-\rho & \rho \end{pmatrix}$$

Starting from $P(y = y_L) = 0.2$, the figure below shows the share of low-income households over a 100-year horizon for each value of ρ.



Explanation:

For all $\rho \in (0, 1)$, the share of low-income households converges to 0.5, which is the unique stationary distribution of this symmetric Markov chain, but higher persistence slows the rate of convergence.

The two edge cases behave differently: $\rho = 0$ causes full state-switching each period (so the share oscillates before settling), while $\rho = 1$ makes both states absorbing, so the initial distribution $P(y_L) = 0.2$ is permanent.

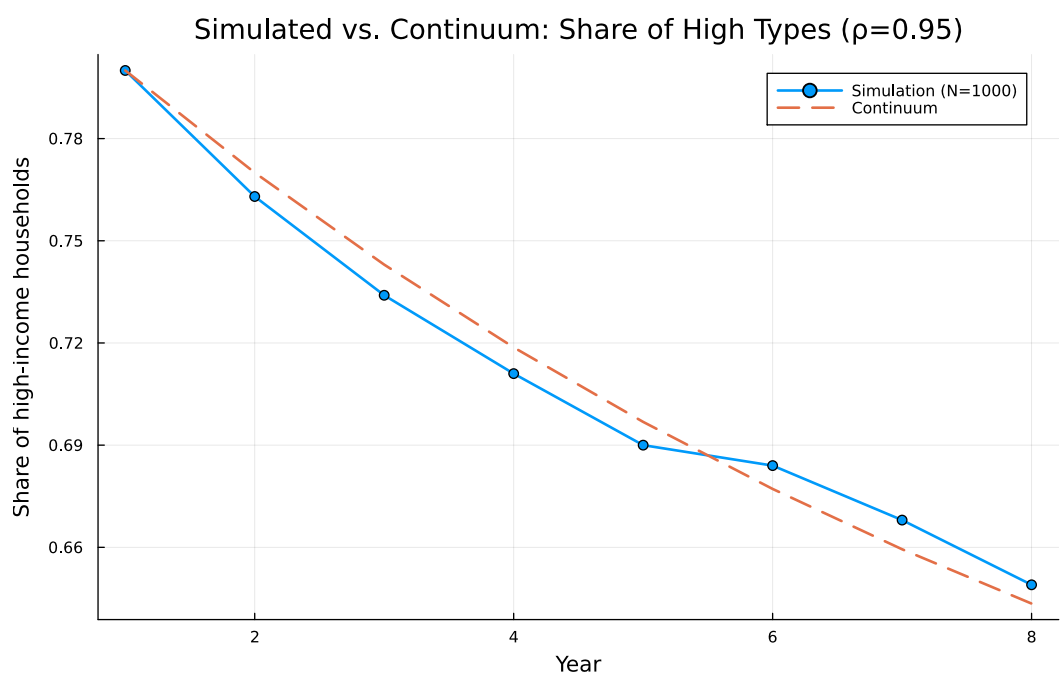
Q2: Volatility vs. Risk

Explanation:

In the stationary distribution, the variance of $\Delta \log y$ equals $(1 - \rho) \cdot (\log 2)^2$, which is maximized at $\rho = 0$ where every household switches state each period.

However, $\rho = 0$ does not generate the most risk for households: with i.i.d. income, bad shocks are transitory and easily self-insured through savings, whereas high persistence (e.g. $\rho = 0.95$) means adverse shocks last many periods, making them far harder to smooth.

Q3: Simulation vs. Continuum



The simulated share of high-income types closely tracks the continuum prediction but does not match it exactly, because a finite sample of 1000 households is subject to sampling variation whereas the continuum result holds by the law of large numbers over a unit mass of agents.